

Table 1
Development Set Results for Task 6.1.1.

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.10	0.03	0.02	0.04	0.02	0.03
				✓	×	×	0.43	0.69	0.50	0.41	0.71	0.52
				×	×	×	0.32	0.13	0.13	0.31	0.29	0.30
				✓	×	×	0.44	0.64	0.51	0.42	0.71	0.52
				×	×	×	0.44	0.26	0.29	0.32	0.39	0.35
				✓	×	×	0.46	0.64	0.52	0.41	0.72	0.53
			✓	×	×	×	0.36	0.08	0.06	0.31	0.27	0.29
				✓	×	×	0.47	0.64	0.53	0.42	0.72	0.53
				×	✓	×	0.49	0.76	0.58	0.49	0.78	0.60
				✓	✓	×	0.48	0.78	0.59	0.47	0.79	0.59
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.48	0.76	0.58	0.48	0.78	0.59
	End	✓	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.50	0.35	0.39	0.51	0.35	0.41
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.51	0.69	0.58	0.51	0.70	0.59
				×	✓	×	0.51	0.76	0.60	0.52	0.78	0.63
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.48	0.76	0.58	0.48	0.78	0.59
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.50	0.78	0.60	0.48	0.80	0.60
				×	✓	×	0.51	0.76	0.60	0.52	0.78	0.63
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
3.2 3B	×	×	×	×	×	×	0.13	0.02	0.02	0.04	0.01	0.02
				✓	×	×	0.45	0.71	0.53	0.43	0.73	0.54
				×	×	×	0.14	0.07	0.04	0.27	0.27	0.27
				✓	×	×	0.47	0.62	0.52	0.41	0.71	0.52
			✓	×	×	×	0.31	0.19	0.20	0.29	0.32	0.30
				✓	×	×	0.45	0.64	0.51	0.41	0.71	0.52
				×	×	×	0.35	0.08	0.06	0.30	0.27	0.29
				✓	×	×	0.48	0.65	0.54	0.43	0.73	0.54
				×	✓	×	0.49	0.76	0.58	0.49	0.78	0.60
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.50	0.78	0.60	0.47	0.79	0.59
	End	✓	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.48	0.76	0.58	0.48	0.78	0.59
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.49	0.77	0.59	0.47	0.79	0.59
				×	✓	×	0.50	0.74	0.59	0.50	0.77	0.61
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
			✓	×	✓	×	0.52	0.63	0.56	0.52	0.62	0.57
				✓	✓	×	0.50	0.73	0.58	0.50	0.76	0.60
				×	×	×	0.49	0.80	0.60	0.51	0.81	0.62
				✓	✓	✓	0.50	0.62	0.54	0.52	0.59	0.56
				×	×	×	0.13	0.02	0.02	0.04	0.01	0.02
				✓	×	×	0.45	0.71	0.53	0.43	0.73	0.54
				×	×	×	0.14	0.07	0.04	0.27	0.27	0.27
				✓	×	×	0.47	0.62	0.52	0.41	0.71	0.52
				×	×	×	0.31	0.19	0.20	0.29	0.32	0.30
				✓	×	×	0.45	0.64	0.51	0.41	0.71	0.52
				×	×	×	0.35	0.08	0.06	0.30	0.27	0.29
				✓	×	×	0.48	0.65	0.54	0.43	0.73	0.54
				×	✓	×	0.49	0.76	0.58	0.49	0.78	0.60
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.50	0.78	0.60	0.47	0.79	0.59
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.48	0.76	0.58	0.48	0.78	0.59
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.49	0.77	0.59	0.47	0.79	0.59
				×	✓	×	0.50	0.74	0.59	0.50	0.77	0.61
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.52	0.63	0.56	0.52	0.62	0.57
				✓	✓	×	0.50	0.73	0.58	0.50	0.76	0.60
Naive	×	×	×	×	✓	×	0.49	0.80	0.60	0.51	0.81	0.62
						✓	0.50	0.62	0.54	0.52	0.59	0.56

A. Extended Results

This appendix holds the complete results for all our evaluated combinations

A.1. Development Evaluation Results

ToGS results are contained in Tables 1 to 4.

Merged results are contained in Tables 5 to 8.

Table 2
Development Set Results for Task 6.1.2.

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.07	0.03	0.02	0.03	0.02	0.02
				✓	×	×	0.37	0.59	0.43	0.34	0.58	0.43
				×	×	×	0.21	0.10	0.10	0.25	0.24	0.24
				✓	×	×	0.37	0.53	0.43	0.33	0.57	0.42
				×	×	×	0.32	0.20	0.22	0.25	0.30	0.28
				✓	×	×	0.38	0.52	0.43	0.33	0.57	0.42
			✓	×	×	×	0.18	0.06	0.04	0.25	0.22	0.23
				✓	×	×	0.39	0.53	0.44	0.34	0.58	0.43
				×	✓	×	0.42	0.66	0.50	0.41	0.66	0.51
				✓	✓	×	0.42	0.67	0.51	0.40	0.66	0.50
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.41	0.65	0.50	0.39	0.64	0.48
	End	✓	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.43	0.31	0.35	0.43	0.29	0.34
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.44	0.60	0.50	0.42	0.58	0.49
				×	✓	×	0.44	0.65	0.51	0.43	0.64	0.52
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
3.2 3B	×	×	×	×	×	×	0.04	0.01	0.01	0.03	0.01	0.02
				✓	×	×	0.38	0.61	0.45	0.35	0.59	0.44
				×	×	×	0.09	0.05	0.03	0.22	0.21	0.21
				✓	×	×	0.40	0.52	0.43	0.33	0.57	0.42
			✓	×	×	×	0.23	0.15	0.15	0.22	0.25	0.24
				✓	×	×	0.38	0.53	0.43	0.33	0.56	0.41
				×	×	×	0.24	0.06	0.04	0.24	0.21	0.22
				✓	×	×	0.41	0.55	0.46	0.34	0.58	0.43
				×	✓	×	0.43	0.67	0.51	0.42	0.67	0.51
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.43	0.67	0.52	0.39	0.65	0.49
	End	✓	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.42	0.65	0.50	0.39	0.64	0.49
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.43	0.67	0.51	0.40	0.66	0.50
				×	✓	×	0.44	0.64	0.51	0.42	0.64	0.50
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
			✓	✓	✓	×	0.45	0.55	0.48	0.43	0.52	0.47
				✓	✓	×	0.43	0.64	0.51	0.42	0.64	0.51
Naive	×	×	×	×	✓	✓	0.50	0.62	0.54	0.52	0.59	0.56

A.2. Test Evaluation Results

ToGS results are contained in Tables 9 to 12.

Merged results are contained in Tables 13 to 15.

References

Table 3
Development Set Results for Task 6.2.1.

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.00	0.01	0.00	0.01	0.01	0.01
				✓	×	×	0.00	0.01	0.00	0.01	0.01	0.01
				×	×	×	0.07	0.09	0.04	0.08	0.11	0.10
				✓	✓	×	0.07	0.09	0.04	0.08	0.11	0.09
			✓	×	×	×	0.05	0.07	0.05	0.06	0.09	0.07
				×	✓	×	0.05	0.07	0.05	0.06	0.09	0.07
				×	×	×	0.04	0.06	0.03	0.07	0.08	0.07
				✓	✓	×	0.04	0.06	0.03	0.07	0.08	0.07
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.00	0.00	0.00	0.00	0.00	0.00
3.2 3B	×	×	×	×	×	×	0.04	0.04	0.03	0.08	0.11	0.09
				✓	✓	×	0.03	0.04	0.03	0.08	0.10	0.09
				×	×	×	0.04	0.05	0.04	0.05	0.07	0.06
				×	✓	×	0.04	0.05	0.04	0.05	0.07	0.06
			✓	×	×	×	0.04	0.05	0.03	0.07	0.09	0.08
				×	×	×	0.04	0.05	0.03	0.07	0.09	0.08
				✓	✓	×	0.04	0.05	0.03	0.07	0.09	0.08
				✓	✓	×	0.04	0.05	0.03	0.07	0.09	0.08

Table 4

Development Set Results for Task 6.2.2.

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.01	0.02	0.01	0.01	0.02	0.01
				✓	×	×	0.02	0.03	0.02	0.01	0.02	0.01
				×	×	×	0.06	0.09	0.05	0.07	0.11	0.09
			✓	×	×	×	0.06	0.09	0.05	0.07	0.10	0.08
				×	×	×	0.07	0.11	0.07	0.06	0.10	0.07
				×	×	×	0.07	0.11	0.07	0.06	0.10	0.07
		✓	×	×	×	×	0.04	0.05	0.03	0.05	0.07	0.06
				×	×	×	0.04	0.05	0.03	0.05	0.07	0.06
				×	×	×	0.01	0.01	0.00	0.01	0.01	0.01
			✓	×	×	×	0.01	0.01	0.00	0.01	0.01	0.01
				×	×	×	0.01	0.01	0.01	0.01	0.01	0.01
				×	×	×	0.01	0.00	0.00	0.01	0.01	0.01
	End	✓	×	×	×	×	0.00	0.00	0.00	0.01	0.00	0.00
				×	×	×	0.01	0.00	0.00	0.01	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.01	0.00	0.00
			✓	×	×	×	0.00	0.00	0.00	0.01	0.00	0.00
				×	×	×	0.06	0.05	0.04	0.07	0.10	0.08
				×	×	×	0.06	0.05	0.04	0.07	0.10	0.08
3.2 3B	×	×	×	×	×	×	0.05	0.07	0.05	0.05	0.09	0.06
				×	×	×	0.05	0.07	0.05	0.05	0.09	0.06
				×	×	×	0.05	0.04	0.03	0.05	0.07	0.06
		✓	×	×	×	×	0.02	0.03	0.02	0.02	0.02	0.02
				×	×	×	0.02	0.03	0.02	0.02	0.02	0.02
				×	×	×	0.00	0.01	0.00	0.01	0.01	0.01
		✓	×	×	×	×	0.00	0.01	0.00	0.01	0.01	0.01
				×	×	×	0.00	0.00	0.00	0.01	0.01	0.01
				×	×	×	0.00	0.00	0.00	0.01	0.01	0.01
	End	✓	×	×	×	×	0.00	0.00	0.00	0.01	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.01	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.01	0.00	0.00

Table 5

Merged Development Set Results for Task 6.1.1, including the results of the merged approaches with Graphwise.

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
NER-EXP-SUB-EL-freq-gaze	∩	3.1 8B	×	✓	×	✓	✓	0.89	0.70	0.76	0.90	0.74	0.81
		3.2 3B	×	✓	×	✓	✓	0.89	0.70	0.76	0.90	0.74	0.81
		End	✓	✓	×	✓	✓	0.90	0.68	0.75	0.90	0.72	0.80
		Naive	×	×	×	×	✓	0.89	0.70	0.76	0.90	0.74	0.81
	∪	3.1 8B	×	✓	×	✓	✓	0.49	0.80	0.60	0.50	0.87	0.64
		3.2 3B	×	✓	×	✓	✓	0.50	0.80	0.60	0.50	0.87	0.64
		End	✓	✓	×	✓	✓	0.52	0.80	0.62	0.55	0.87	0.67
		Naive	×	×	×	×	✓	0.50	0.81	0.61	0.53	0.88	0.66
NER-EXP-SUB	∪	3.1 8B	×	✓	×	✓	✓	0.50	0.81	0.61	0.51	0.87	0.64
		3.2 3B	×	✓	×	✓	✓	0.50	0.80	0.61	0.51	0.87	0.64
		End	✓	✓	×	✓	✓	0.52	0.81	0.62	0.55	0.87	0.68
		Naive	×	×	×	×	✓	0.51	0.82	0.62	0.53	0.88	0.66

Table 6

Merged Development Set Results for Task 6.1.2, including the results of the merged approaches with Graphwise.

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
NER-EXP-SUB-EL-freq-gaze	$\cap$	3.1 8B	×	✓	×	✓	✓	0.86	0.67	0.74	0.85	0.70	0.77
		3.2 3B	×	✓	×	✓	✓	0.86	0.67	0.74	0.85	0.70	0.77
		End	✓	✓	×	✓	✓	0.87	0.65	0.73	0.86	0.69	0.76
		Naive	×	×	×	×	✓	0.86	0.67	0.74	0.85	0.70	0.77
	$\cup$	3.1 8B	×	✓	×	✓	✓	0.43	0.69	0.52	0.42	0.72	0.53
		3.2 3B	×	✓	×	✓	✓	0.44	0.69	0.53	0.42	0.72	0.53
		End	✓	✓	×	✓	✓	0.45	0.69	0.53	0.46	0.72	0.56
		Naive	×	×	×	×	✓	0.44	0.70	0.53	0.44	0.73	0.55
	NER-EXP-SUB	3.1 8B	×	✓	×	✓	✓	0.24	0.39	0.30	0.22	0.38	0.28
		3.2 3B	×	✓	×	✓	✓	0.25	0.39	0.30	0.22	0.38	0.28
		End	✓	✓	×	✓	✓	0.24	0.37	0.29	0.23	0.37	0.29
		Naive	×	×	×	×	✓	0.25	0.40	0.30	0.23	0.39	0.29

Table 7

Merged Development Set Results for Task 6.2.1, including the results of the merged approaches with Graphwise.

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
gpt-5.shot-gold	$\cap$	3.2 3B	×	×	×	✓	×	0.17	0.03	0.05	0.69	0.09	0.17
	$\cup$	3.2 3B	×	×	×	✓	×	0.24	0.59	0.31	0.24	0.64	0.35

Table 8

Merged Development Set Results for Task 6.2.2, including the results of the merged approaches with Graphwise.

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
gpt-5.shot-gold	$\cap$	3.2 3B	×	×	×	✓	×	0.22	0.04	0.06	0.62	0.08	0.15
	$\cup$	3.2 3B	×	×	×	✓	×	0.23	0.60	0.30	0.21	0.63	0.31

Table 9

Test Set Results for Task 6.1.1

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.10	0.03	0.02	0.03	0.02	0.02
				✓	×	×	0.28	0.42	0.31	0.29	0.48	0.36
			✓	×	×	×	0.37	0.22	0.25	0.31	0.37	0.34
				✓	×	×	0.34	0.45	0.37	0.35	0.58	0.44
			✓	×	×	×	0.20	0.07	0.05	0.32	0.27	0.29
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
		✓	×	×	×	×	0.31	0.45	0.35	0.34	0.53	0.42
				✓	×	×	0.31	0.45	0.35	0.33	0.53	0.41
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.33	0.45	0.37	0.33	0.53	0.41
3.2 3B	×	×	×	×	×	×	0.05	0.02	0.01	0.03	0.01	0.02
				✓	×	×	0.28	0.42	0.31	0.30	0.49	0.37
				✓	×	×	0.32	0.39	0.33	0.34	0.55	0.42
				×	×	×	0.32	0.17	0.19	0.29	0.32	0.30
			✓	×	×	×	0.33	0.44	0.36	0.34	0.57	0.43
				✓	×	×	0.11	0.07	0.04	0.31	0.27	0.29
				✓	×	×	0.32	0.39	0.33	0.34	0.55	0.42
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
		✓	×	×	×	×	0.30	0.44	0.34	0.34	0.53	0.42
				✓	×	×	0.33	0.45	0.37	0.33	0.53	0.41
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.33	0.45	0.36	0.33	0.53	0.40
Naive	×	×	×	×	✓	×	0.31	0.46	0.36	0.35	0.54	0.43
						✓	0.28	0.28	0.27	0.33	0.34	0.33

Table 10

Test Set Results for Task 6.1.2

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.07	0.02	0.01	0.02	0.01	0.02
				✓	×	×	0.19	0.28	0.21	0.20	0.33	0.25
			✓	×	×	×	0.19	0.12	0.13	0.19	0.23	0.21
				✓	×	×	0.20	0.26	0.21	0.21	0.35	0.27
				✓	×	×	0.09	0.05	0.03	0.22	0.18	0.20
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
		✓	×	×	✓	×	0.22	0.31	0.24	0.24	0.38	0.30
				✓	✓	×	0.22	0.31	0.25	0.23	0.37	0.28
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.24	0.32	0.26	0.24	0.38	0.29
3.2 3B	×	×	×	×	×	×	0.04	0.01	0.01	0.02	0.01	0.01
				✓	×	×	0.20	0.29	0.22	0.21	0.35	0.26
				✓	✓	×	0.22	0.26	0.22	0.22	0.36	0.28
			✓	×	×	×	0.16	0.09	0.10	0.18	0.20	0.19
				✓	✓	×	0.21	0.27	0.22	0.22	0.36	0.27
				×	×	×	0.03	0.04	0.02	0.21	0.18	0.19
				✓	✓	×	0.21	0.25	0.22	0.22	0.35	0.27
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.21	0.30	0.23	0.24	0.37	0.29
				✓	✓	×	0.23	0.30	0.25	0.23	0.37	0.28
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.23	0.30	0.25	0.23	0.37	0.28
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.23	0.30	0.25	0.23	0.37	0.28
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.23	0.30	0.25	0.23	0.37	0.28
Naive	×	×	×	×	✓	×	0.22	0.32	0.25	0.25	0.38	0.30
						✓	0.25	0.25	0.24	0.31	0.31	0.31

Table 11

Test Set Results for Task 6.2.1

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.00	0.01	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.01	0.00	0.00	0.00	0.00
				×	×	×	0.11	0.08	0.06	0.11	0.12	0.11
				✓	×	×	0.11	0.08	0.06	0.11	0.12	0.11
				×	×	×	0.09	0.07	0.06	0.07	0.09	0.08
				✓	×	×	0.09	0.07	0.06	0.07	0.09	0.08
			✓	×	×	×	0.04	0.05	0.02	0.10	0.10	0.10
				✓	×	×	0.04	0.05	0.02	0.10	0.10	0.10
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
3.2 3B	×	×	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				×	×	×	0.08	0.06	0.05	0.12	0.11	0.11
				✓	×	×	0.08	0.06	0.05	0.12	0.11	0.11
			✓	×	×	×	0.06	0.05	0.05	0.06	0.07	0.07
				✓	×	×	0.06	0.05	0.05	0.06	0.07	0.07
				×	×	×	0.04	0.05	0.02	0.11	0.10	0.10
				✓	×	×	0.04	0.05	0.02	0.11	0.10	0.10
			×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00

Table 12

Test Set Results for Task 6.2.2

Model	Beams	NED FT	RAG	LoRA	Naive	Filter	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
3.1 8B	×	×	×	×	×	×	0.00	0.02	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.02	0.00	0.00	0.00	0.00
				×	×	×	0.05	0.03	0.03	0.06	0.07	0.06
			✓	✓	×	×	0.05	0.03	0.03	0.06	0.07	0.06
				×	×	×	0.05	0.04	0.03	0.04	0.05	0.04
				✓	×	×	0.05	0.04	0.03	0.04	0.05	0.04
		✓	×	×	×	×	0.02	0.03	0.01	0.05	0.06	0.06
				✓	×	×	0.02	0.03	0.01	0.05	0.06	0.06
				×	×	×	0.00	0.01	0.00	0.00	0.00	0.00
			✓	×	×	×	0.00	0.01	0.00	0.00	0.00	0.00
				×	×	×	0.01	0.01	0.01	0.01	0.01	0.01
				✓	✓	×	0.02	0.00	0.00	0.00	0.00	0.00
3.2 3B	×	×	×	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				×	×	×	0.03	0.04	0.03	0.06	0.07	0.06
			✓	✓	×	×	0.03	0.04	0.03	0.06	0.07	0.06
				×	×	×	0.03	0.04	0.03	0.04	0.05	0.04
				✓	×	×	0.03	0.04	0.03	0.04	0.05	0.04
		✓	×	×	×	×	0.02	0.03	0.01	0.05	0.05	0.05
				✓	×	×	0.02	0.03	0.01	0.05	0.05	0.05
				×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
			✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00
				✓	✓	×	0.01	0.01	0.01	0.01	0.01	0.01
				×	×	×	0.02	0.01	0.01	0.01	0.01	0.01
			✓	✓	✓	×	0.00	0.00	0.00	0.00	0.00	0.00

Table 13

Merged Test Set Results for Task 6.1.1

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
ENS-EXP-SUB-1	∪	3.2 3B	×	✓	×	✓	✓	0.48	0.75	0.58	0.50	0.83	0.63
		Naive	×	×	×	×	✓	0.48	0.75	0.57	0.52	0.84	0.64
ENS-EXP-SUB-EL-freq-gaze	∩	3.2 3B	×	✓	×	✓	✓	0.75	0.42	0.52	0.89	0.51	0.65
		Naive	×	×	×	×	✓	0.75	0.42	0.52	0.89	0.51	0.65
	∪	3.2 3B	×	✓	×	✓	✓	0.48	0.75	0.58	0.50	0.83	0.62
		Naive	×	×	×	×	✓	0.48	0.75	0.57	0.52	0.83	0.64
NER-EXP-SUB-1	∪	3.2 3B	×	✓	×	✓	✓	0.48	0.75	0.58	0.51	0.84	0.63
		Naive	×	×	×	×	✓	0.48	0.75	0.58	0.53	0.84	0.65
NER-EXP-SUB-EL-freq-gaze	∩	3.2 3B	×	✓	×	✓	✓	0.74	0.43	0.52	0.87	0.51	0.65
		Naive	×	×	×	×	✓	0.74	0.43	0.52	0.87	0.51	0.65
	∪	3.2 3B	×	✓	×	✓	✓	0.48	0.75	0.58	0.51	0.83	0.63
		Naive	×	×	×	×	✓	0.48	0.75	0.57	0.53	0.84	0.65
gpt-5.shot-gold	∪	3.2 3B	×	✓	×	✓	✓	0.34	0.46	0.37	0.33	0.54	0.41
		Naive	×	×	×	×	✓	0.31	0.46	0.36	0.36	0.54	0.43

Table 14
Merged Test Set Results for Task 6.1.2

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
ENS-EXP-SUB-1	$\cup$	3.2 3B	×	✓	×	✓	✓	0.12	0.18	0.14	0.13	0.21	0.16
		Naive	×	×	×	×	✓	0.11	0.18	0.14	0.13	0.21	0.16
ENS-EXP-SUB-EL-freq-gaze	$\cap$	3.2 3B	×	✓	×	✓	✓	0.63	0.36	0.45	0.77	0.44	0.56
		Naive	×	×	×	×	✓	0.63	0.36	0.45	0.77	0.44	0.56
	$\cup$	3.2 3B	×	✓	×	✓	✓	0.27	0.42	0.32	0.30	0.50	0.38
		Naive	×	×	×	×	✓	0.26	0.42	0.32	0.31	0.50	0.39
NER-EXP-SUB-1	$\cup$	3.2 3B	×	✓	×	✓	✓	0.12	0.18	0.14	0.13	0.21	0.16
		Naive	×	×	×	×	✓	0.11	0.18	0.13	0.13	0.20	0.16
NER-EXP-SUB-EL-freq-gaze	$\cap$	3.2 3B	×	✓	×	✓	✓	0.63	0.37	0.45	0.76	0.45	0.56
		Naive	×	×	×	×	✓	0.63	0.37	0.45	0.76	0.45	0.56
	$\cup$	3.2 3B	×	✓	×	✓	✓	0.27	0.43	0.33	0.31	0.50	0.38
		Naive	×	×	×	×	✓	0.27	0.43	0.32	0.32	0.50	0.39
gpt-5.shot-gold	$\cup$	3.2 3B	×	✓	×	✓	✓	0.24	0.32	0.26	0.24	0.38	0.29
		Naive	×	×	×	×	✓	0.22	0.32	0.25	0.25	0.38	0.30

Table 15
Merged Test Set Results for Task 6.2.2

Graphwise	Set	Model	Beams	NED FT	RAG	LoRA	Naive	P	R	F_1	P_{micro}	R_{micro}	$F_{1,micro}$
ENS-EXP-SUB-1	$\cup$	3.2 3B	×	×	×	✓	×	0.03	0.04	0.03	0.06	0.07	0.06
ENS-EXP-SUB-EL-freq-gaze	$\cup$	3.2 3B	×	×	×	✓	×	0.03	0.04	0.03	0.06	0.07	0.06
NER-EXP-SUB-1	$\cup$	3.2 3B	×	×	×	✓	×	0.03	0.04	0.03	0.06	0.07	0.06
NER-EXP-SUB-EL-freq-gaze	$\cup$	3.2 3B	×	×	×	✓	×	0.03	0.04	0.03	0.06	0.07	0.06
gpt-5.shot-gold	$\cap$	3.2 3B	×	×	×	✓	×	0.05	0.01	0.01	0.34	0.02	0.04
	$\cup$	3.2 3B	×	×	×	✓	×	0.07	0.20	0.10	0.09	0.22	0.12